

An aerial photograph showing a flooded area in Nepal. A river with muddy, fast-moving water flows through the scene. A bridge is partially submerged, with debris caught in the rapids. On the left bank, a group of people is gathered near some buildings. In the background, there are colorful buildings, including one with a sign that says "HOTEL RIVERVIEW". The surrounding area is lush with green trees and vegetation.

HOW CAN WE BETTER PREPARE **NEPAL** FOR **FLOODS**?

Flood resistant building code for Nepal



Help to develop a **new guide within the building code for Nepal** that accounts for **flood risks**! Nepal already has a building code focused on seismic hazard (earthquake), which has been revised following the devastating earthquakes in 2015. Yet, there are no international, national, or local guidelines help to deal with big flood events in the southern part of Nepal, the plains (**Teraï region**) in combination with earthquake risks. We need to understand how to deal with **multihazard-risks** in the built environment.

Remember the big recent floods from Pakistan? This is also a wake-up call for Nepal! Help Nepal get prepared for what is coming and becoming more **climate adaptive**!

Flood resistant building code for Nepal – join the team!

Use your **internship**, your **bachelor or master thesis**, or your time as a **young professional**, or **early career researcher** to make a difference!

We are building a team with extra support from researchers of the **NHSRP**, **University of Twente ITC**, **Oxford Brookes Cendep**, **CRAterre**, **Tribhuvan University**, **University of Sydney** and **CRS** to guide you in your research. Which challenge suits you best?



Challenge: Lessons from vernacular architecture



Research question

What lessons can be learned from flood resistance of vernacular architecture in the Terai area?

Description

We do not only need a **profound literature study**, but an analysis of the local building culture and **vernacular architecture** of this flood prone regions of Nepal. This challenge might require you to go to Nepal and experiment and/ or adapt existing tools.

Proposed methodology

- A. Field surveys to document the practices adapted to the flood risk, including technical data collection, but also interviews with local stakeholders, participatory workshops.
- B. Remotely, collaborate with local partners to co-define the objectives and content of the surveys and synthesize the data collected.

Potential supervisors

Julien Hosta, Olivier Moles, Dr. Eefje Hendriks, Charles Parrack,

Suggested background

Architecture, construction engineering, structural engineering

Challenge: Vernacular flood resistant construction measures

Research question

What vernacular flood-resistant construction principles are appropriate and effective to reduce risks in flood-prone rural areas in Nepal?

Description

One of the most promising ways to develop a building code of Nepal is mapping what is already been used by communities living in flood prone areas. Their own designs likely have proven their resistance to floods over the years and have been passed on over generations. It is also likely that they use locally available materials and feasible techniques. Currently, there is not documentation of what is already there.

Proposed methodology

- A. **Prepare assessment tools** to document vernacular flood resilient housing designs.
- B. Travel to Nepal, to the Terai area and **document vernacular flood resilient housing designs** by systematically making photos and measurements, or collaborate with the team for NHSRP to collect data.
- C. Translate the collected information of the **designs into drawings** floor plans, elevations, connections and sections (optional 3D models).
- D. Making a catalogue with **current flood resilient construction** techniques in Nepal describing typologies, and identifying best practices, gaps, and challenges. **Evaluate the designs** on criteria such as flood resistance, durability, sustainability, circularity, cost, availability of local materials.
- E. **Discuss your findings** with NHSRP, humanitarian organizations, civil engineers and using literature from scientific studies.

Potential supervisors

Julien Hosta, Olivier Moles, Eefje Hendriks, Charles Parrack,

Suggested background

Architecture, construction engineering, structural engineering



Challenge: Building codes for flood resilience

Research question

What is the scientific consensus on the effectivity of flood-resistant construction principles for low-cost housing?

Description

Building codes exist in many countries, both developed and less developed regions for flood-resistant construction. In order to propose a new building code for Nepal it will be useful to compare different approaches to flood resistant construction.

The first step is data gathering: to gather together in one place as many existing codes and guidance as can be found and categorise them.

The next step is to compare approach to flood resistance in the documents, and propose which are suitable and appropriate for Nepal

Finally, to analyse the scientific basis of the flood resistance, and whether there are gaps in the science which need to be addressed.

Proposed methodology

- A. Systematic literature review of all floods resilient building codes in context of Nepal in [this sheet](#).
- B. Analysis of similarities and differences in existing building codes.
- C. Analysis of the scientific origin of the recommendations in building codes and identifying gaps for further scientific testing.

Potential supervisors

Charles Parrack, Eefje Hendriks, Julien Hosta, Olivier Moles

Suggested background

Architecture, construction engineering, structural engineering

Challenge: Key-messages for flood resistance

Research question

What technical guidelines can effectively enable vulnerable communities to become flood resistant in Nepal?

Description

Messages to construct flood resistance are commonly shared in the form of images and pamphlets by governmental departments and humanitarian organizations. For the development of a new building code, it is crucial to map what is already there inside Nepal and in neighboring countries. In this challenge we aim to explore differences and commonalities and identify the value of these messages. At this moment there is very limited scientific base for technical construction guidelines that account for both earthquakes and floods. What guidelines can be trusted?

Proposed methodology

- Collect messages and images** that have been used in **Nepal** to construct flood resistant and earthquake resistant housing and communities.
- Make a compilation** of the images following [this example](#).
- Collect messages and images** that have been used **outside Nepal** to construct flood and earthquake resistant housing, for example from **neighboring countries** Bangladesh, India, China, and Pakistan.
- Make another compilation** of these messages.
- Evaluate the scientific value of key messages** for the Terai area in Nepal in conversation with NHSRP, humanitarian organizations, civil engineers and using literature from scientific studies. Identify needs for further research. What would you recommend?

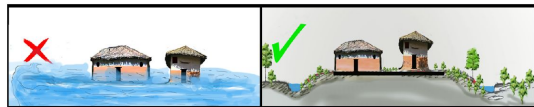
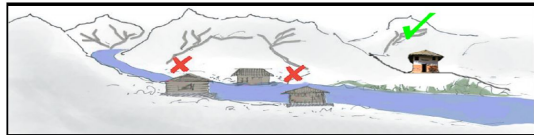
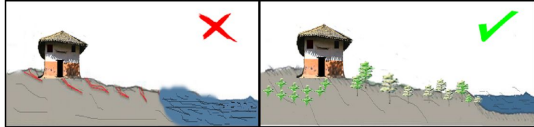
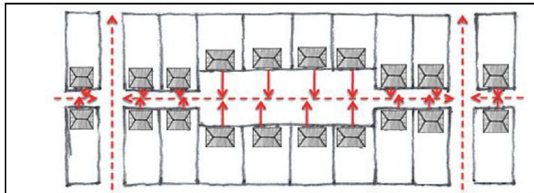
Potential supervisors

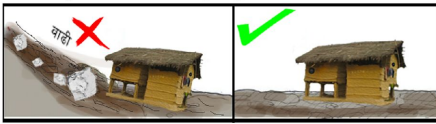
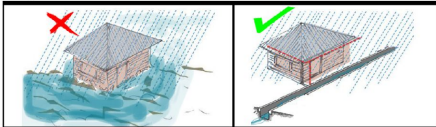
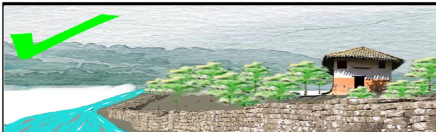
Eefje Hendriks, Charles Parrack, Minar Thapa Magar, Inu Pradhan Salike, Julien Hosta, Olivier Moles

Suggested background

Communication science, behavioural science, Humanitarian assistance, development studies, disaster studies

घर तथा बस्तीमा बाढी, पहिरोको जोखिम कम गर्ने केही उपायहरू :-

- घर बनाउँदा पानी नजम्ने तथा बाढी नपर्ने र सकभर अग्ला स्थान छान्ने ।

- घर तथा बस्ती अचानक र तिब्रगतिको बाढीको बहाव हुनसक्ने क्षेत्रभन्दा टाढा हुनुपर्छ ।

- घर तथा बस्तीको वरिपरि भू-क्षय हुनवाट जोगाउन बोटबिरुवा रोप्नुपर्छ ।

- बसोबास क्षेत्र बिचमा खाली ठाउँ छोडेर बाहिरपट्टी बनाएको हुनुपर्छ ताकी स्थानान्तरण र उद्धार गर्न सजिलै पहुँच पुगोस् ।

- घर तथा बस्तीको स्थान वर्षाको समयमा बाढी पहिरो आउनसक्ने जोखिम भएको स्थान भन्दा टाढा हुनुपर्छ ।

- घर तथा बस्तीको लागि बाढी आईरहेको समयमा पहिरो आएर पुरीनसक्ने सम्भावना नहुने समथर जमिन उपयुक्त हुन्छ ।

- घरको वरिपरि आवश्यकता अनुसार नाला काटेर पानीको निकास सहज बनाउनु पर्छ ।

- बस्तीमा बाढी पस्नवाट रोक्न उपयुक्त संरचनाहरू निर्माण गरि जैविक प्रविधिको प्रयोग गर्नुपर्दछ ।


Challenge: Co-create safety measures with communities at risk

Research question

What are effective ways to develop and introduce new building regulations into flood-prone rural communities in Nepal?

Description

Currently, there are no governmental or humanitarian reconstruction assistance approaches that are personalized to the situation, motivations, abilities, and opportunities of individual households at risk. In this challenge, we aim to develop a method to adapt reconstruction assistance to individuals in multi-hazard situations throughout the reconstruction process, enabling vulnerable communities to make informed construction decisions.

Often building codes and spatial plans are developed without engagement of these communities. However, the introduction of new building codes and spatial plans might require another way of living and land use. Therefore, communities should be granted possibilities to co-create the building code and spatial plans. Based on existing co-creation strategies, a method is needed to engage different types of households in multi-hazard situations effectively in spatial policy making, and housing designs to enhance resilience. This method should facilitate governmental and humanitarian stakeholders in Nepal their interaction with communities living in the flood prone areas.

Proposed methodology

- A. **Evaluate theories** about **co-creating methods** for the new building codes / spatial plans.
- B. Interview humanitarian and governmental actors about their **professional experience with community engagement** and analyze their suggestions.
- C. Point out **differences and commonalities** in the strategies and evaluate their importance for different spatial dimensions, such as varying flood and earthquake risks.[E(1)]
- D. **Evaluate the different engagement strategies** on criteria such as effectiveness, inclusion, participation.
- E. **Designing an engagement strategy**[E(2)] **or method** for the Terai area in Nepal with NHSRP. Specify adaptability of the method to individuals and communities in multi-hazard situations.

Potential supervisors

Eefje Hendriks, Charles Parrack, Minar Thapa Magar, Inu Pradhan Salike, Julien Hosta, Olivier Moles

Suggested background

Communication science, behavioural science, Humanitarian assistance, development studies, disaster studies, anthropology, social studies



Challenge: Assess multi-hazard loss and damage

Research question

What are effective ways to assess multi-hazard loss and damage in Nepal?

Description

The primary hazard of Nepal is earthquake, but Nepal suffers yearly with the occurrence of disasters related to multiple hazards like monsoon hazard (heavy rainfall, flood, inundation, landslide), lightning, windstorm, fire, and many more. Most of them associated with the climate change.

The **frequency of occurrence, the extent and the volume** of damages and losses due to the disasters demand different level of assessment, intervention and recovery mechanism. Different assessment tools and methodology have been studied and applied after Gorkha Earthquake 2015 also. However, gaps have been felt for the timely selection of the best possible tool and its implementation in the field after any disaster that is efficient in terms of time, cost as well as quality.

Proposed methodology

- A. Compare existing damage assessment tools** including IRA, MIRA, CSDA, DaLA, PDNA, PDNA lite, and others.
- B. Develop a matrix of the different assessment tools** (cost, applicability, types of disasters,)- This will assist selection decision of a particular tool/methodology depending upon the extent and volume of damage.
- C. Analyse novel tools** that could be used to assess damage, including the use of unmanned air vehicles (drones) and remote sensing (satellites).
- D. Develop a new multi-hazard damage assessment** applicable to multi-hazard situations, including definitions of climate change related damage.
- E. Pilot the assessment tool** in multiple communities/municipalities affected by a disaster
- F. Evaluate and translate the damage assessment tools** into KoBo for digital data collection.

Potential supervisors

Rabin Chaulagain, Eefje Hendriks, Charles Parrack, Julien Hosta, Olivier Moles

Suggested background

Architecture, construction engineering, structural engineering, DRRM

Challenge: Assess vulnerability of flood-prone settlements

Research question

What interventions effectively address vulnerability of flood-prone settlements in Nepal?

Description

Each year during the monsoon, the southern flat area of Nepal gets flooded. In this area there are multiple small and medium size settlements that largely dependent on agriculture. The water from the melting glaciers is collected in different catchments and causes landslides and severe inundations. On top of earthquake risks, the cascade of hazards or non-related hazards, make this a challenging study area. People should ideally be protected for all risks, yet the focus has been primarily on earthquake risks. At the moment, a team of researchers in collaboration with the government and humanitarian organizations is investing in the development of flood-resistant housing that also account for earthquake resistance. However, it is crucial to look at the multi-hazard risks and vulnerability of the settlements. Which settlements are at great risk of being flooded (looking at once in 5, 10, 50, and 100 year flood events)? How can we determine which settlements can be protected and what interventions are effective? Based on a comparison of earlier methods, we aim to develop a new assessment method to select settlements and interventions.

As a first step in this assignment, evaluate existing flood risk maps that have been made and distinguish the different remote sensing approaches that have been used. What is the most appropriate method to assess the vulnerability of communities? Use one of the tools we use within ITC to identify potential spatial interventions to protect communities, for example open LISEM, in collaboration of Bastian van den Bout. Based on your analysis of flood risk maps, try to define preliminary guidelines for safe settlement development.

Proposed methodology

- A. Evaluate **flood risk analysis maps** that have been made for Nepal, categorize their assessment approaches, and try to access the different datasets and maps that have been used.
- B. Compare the different approaches on suitability to assess vulnerability of settlements. s.
- C. Conceptualize how **vulnerability of communities** can be best presented using data and maps.
- D. Quantify the **vulnerability** of flood-prone settlements in Nepal in detail.
- E. Use one of the approaches to **define interventions and guidelines** to enhance the flood resilience of the settlements at risk.

Potential supervisors

Eefje Hendriks e.Hendriks@utwente.nl

Ashok Dahal a.dahal@utwente.nl

Janneke Ettema j.ettema@utwente.nl

Bastian van der Bout b.vandenbout@utwente.nl

Minar Thapa Magar minar@hrrpnepal.org

Suggested background

Architecture, construction
engineering, structural
engineering, spatial engineering,
GIS



TEAM



UNIVERSITY
OF TWENTE.



Minar Thapa Magar
NHSRP



Rabin Chaulagain
NHSRP



Eefje Hendriks
University Twente, ITC



Julien Hosta
CRAtterre



Oliver Moles
CRAtterre



Charles Parrack
CENDEP Oxford Brookes



Inu Pradhan Salike
Tribhuvan University,
Institute of
Engineering(IOE)



Jamie Richardson
CRS



Aaron Opdyke
University of Sydney



Ashok Dahal
University Twente,
ITC



Janneke Ettema
University Twente,
ITC



Bastian van der Bout
University Twente,
ITC

Background and history

Background

The topography of the Terai is almost flat with gentle slopes towards the south. The elevation of the Terai ranges from 65m to 300m from mean sea level with varying widths of 20 km to 45 km. The climate of the Terai is subtropical with average temperature of 25°C. The rainfall of the Terai varies from 1,200 mm to 3,000 mm per annum with occasional showers and cloudbursts. All the river systems flow towards the flat plain of terai from higher mountains and middle hills. During the monsoon months from June to September, all these rivers are in spate with bank-full discharges and cause flooding and inundation in several parts of the Terai. The problems of flooding and inundation in the Terai are more critical due to change in climate in general and change in the rainfall pattern/ intensity in particular.

In a single year 2020, flood and landslide affected 40 districts across the country especially in Terai and hilly regions due to heavy and continuous rain during the monsoon season. A total of 360 people (DRR Portal) lost their lives. According to the NRCS initial rapid assessment (IRA) report, 23,478 families (117,390 people) were affected due to the disaster. The flood inundation is less accessed and mostly occurs in the flat plain of Nepal resulting in losses of life and increasing hardship to the life of affected population. Hence the structure resilient for flood and inundation helps to cope people during the inundation period.

This effort for the compilation of flood resilience design of houses is to explore the best practices and best learnings from the partners and proposed a few flood resilience design options to NDRRMA. The future direction is to update the monsoon Karya Bhidi, adding more clarity on flood inundation impact and support for reconstruction and recovery.

Historic Flood Inundation effect in Terai

Past Flood Events in the Terai Historical evidence shows that the floods of 1785, 1787, 1793, 1806, 1867, and 1871 in the 19th century were major ones. The flood historical data clearly indicates that the 20th century is going to be more challenging mainly due to climate change impact.

During the 1871 floods, rainfall of 483 mm in 36 hours was recorded causing heavy loss of lives and property in Nepal's Terai and India. In 1883, about 2,300 km² of area situated at the northern part of the railway line in India was flooded severely. Thousands of houses collapsed and crops of some 1,000 villages were almost ruined. In 1902, excessive rainfall along the lower reaches of the Himalaya had caused heavy floods in the Nepal Terai. In 1926, heavy floods occurred in the Bagmati River in the Terai and 75 percent of Bhada (harvested at mid September) crops and 50 percent of Aghani (harvested in late November) crops were damaged completely in the Terai and adjoining Indian Territory. In 1934, the floods of the Bagmati River had caused it to shift its course towards the right in India. In 1935, severe floods and inundation again occurred in the Terai and India. The Sitamani railway embankment in India was breached by flood. In recent times, heavy floods were observed in the Nepal Terai and northern India. The floods of 1954, 1956, 1958, 1960, 1962, 1981, 1987, 1993, 1998, 2002, 2004, 2008, 2013, 2019, 2020, 2021 are past events of floods.

Additional information

References

Shelter Response Profiles:

<https://www.sheltercluster.org/promoting-safer-building-working-group/library/shelter-response-profiles>

Local building cultures for sustainable & resilient habitats. Examples of local good practices and technical solutions: <https://craterre.hypotheses.org/1774>

An online course on participatory assessment of local building practices:

<https://www.fap-learning-lab.org/an-online-course-on-participatory-assessment-of-local-building-practices>

Collaboration

We aim to work together to support Nepal in becoming flood resilient. We work together through open collaborations and co-creating. The academic partners are offering their expertise, time and network primarily to address needs of Nepal. Students or young professionals can collaborate directly with the partners involved and will be considered part of the team.